

I would investigate few-label industrial defect inspection with a photonic reservoir

A measurable 12-week path from public data to a conditional hardware test

REAL INDUSTRIAL NEED

Scarce expert labels

Quality inspection needs useful accuracy from small labelled sets.

PHOTONIC FIT

Fixed nonlinear map

Interference transforms compact phase-encoded image features.

12-WEEK TEST

Decision-ready result

Public data · matched controls simulator → conditional QPU run

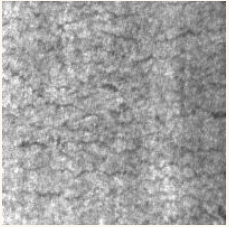
COMPLETED DE-RISKING PILOT

Q1 0.8708 macro-F1 · 48 labels per class

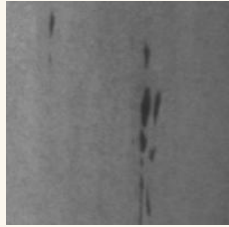
Useful signal; no advantage yet

Evidence informs the next test; it does not support a quantum-advantage claim.

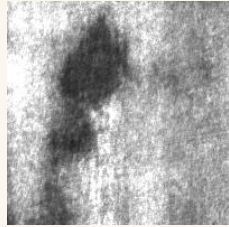
Why this is a good 12-week feasibility study



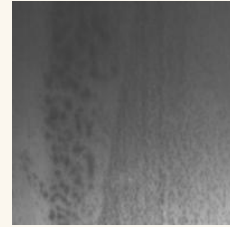
Crazing



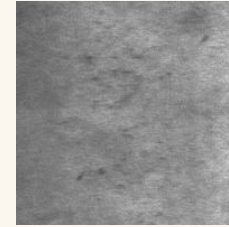
Inclusion



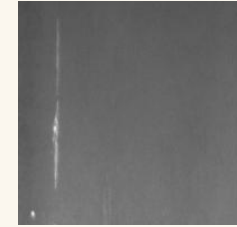
Patches



Pitted surface



Rolled-in scale



Scratches

REAL NEED

Labels are expensive

The target is useful accuracy from
12 · 24 · 48 labels per class.

Sample efficiency is measurable.

TRACTABLE BENCHMARK

1,800 images

Six balanced defect classes
300 images per class

Public data · macro-F1 target

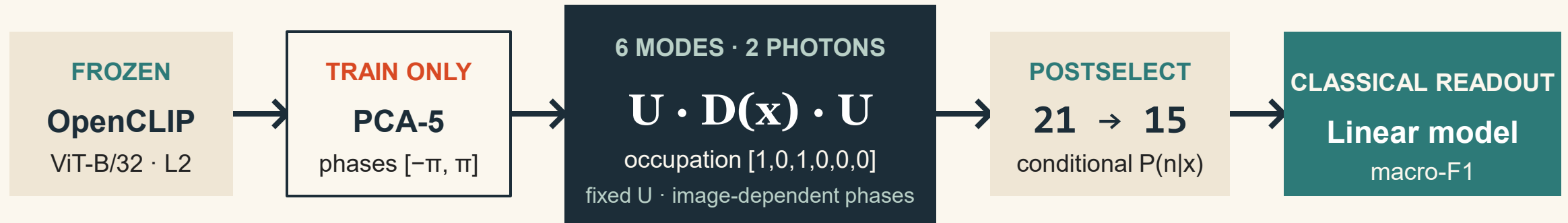
LEAKAGE-SAFE EVIDENCE

One duplicate group

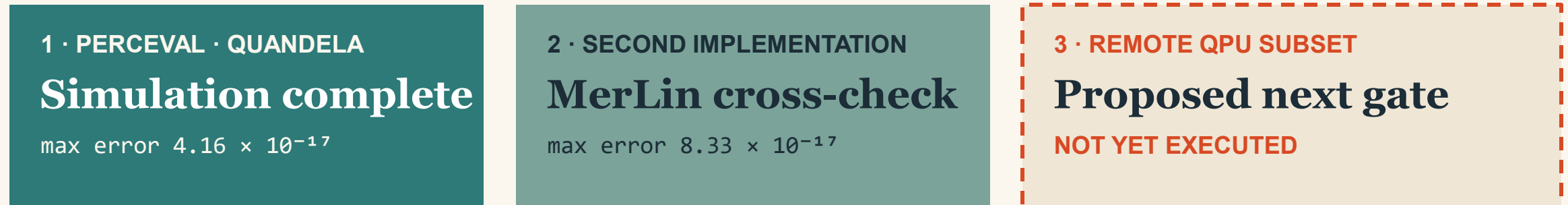
Pa_101 · Pa_105 never cross splits.
Fold 0: 60 test images per class.

Real need · data · decision metric

A small reservoir fits Quandela's path to hardware



$$U(x) = UD(x)U, \quad \phi_Q(x) = \text{Normalize}(\{P(\mathbf{n} | x)\}_{\mathbf{n} \in \mathcal{C}})$$



Source: experiment_protocol.md · verification.json

The feasibility question is deliberately falsifiable

AT THE SAME COMPRESSED INPUT AND 15-D OUTPUT

Can Q1 beat matched RFF, close the ELM / RBF gap, or show practical resource value?

PAIRED PROTOCOL

12 · 24 · 48

total labels per class

Subset seeds 11 · 22 · 33

Map seeds 101 · 202 · 303

Validation macro-F1 only

One test evaluation

per selected pipeline

MAP SEEDS AVERAGED—NEVER PICKED

SEVEN FAMILIES · SAME TEST GATE

B1 Full CLIP + linear

B2 Full CLIP + RBF

B3 PCA-5 + linear

B4 PCA-5 + RBF

B5 RFF-15

B6 ELM-15

Q1-15

CONTINUE GATE

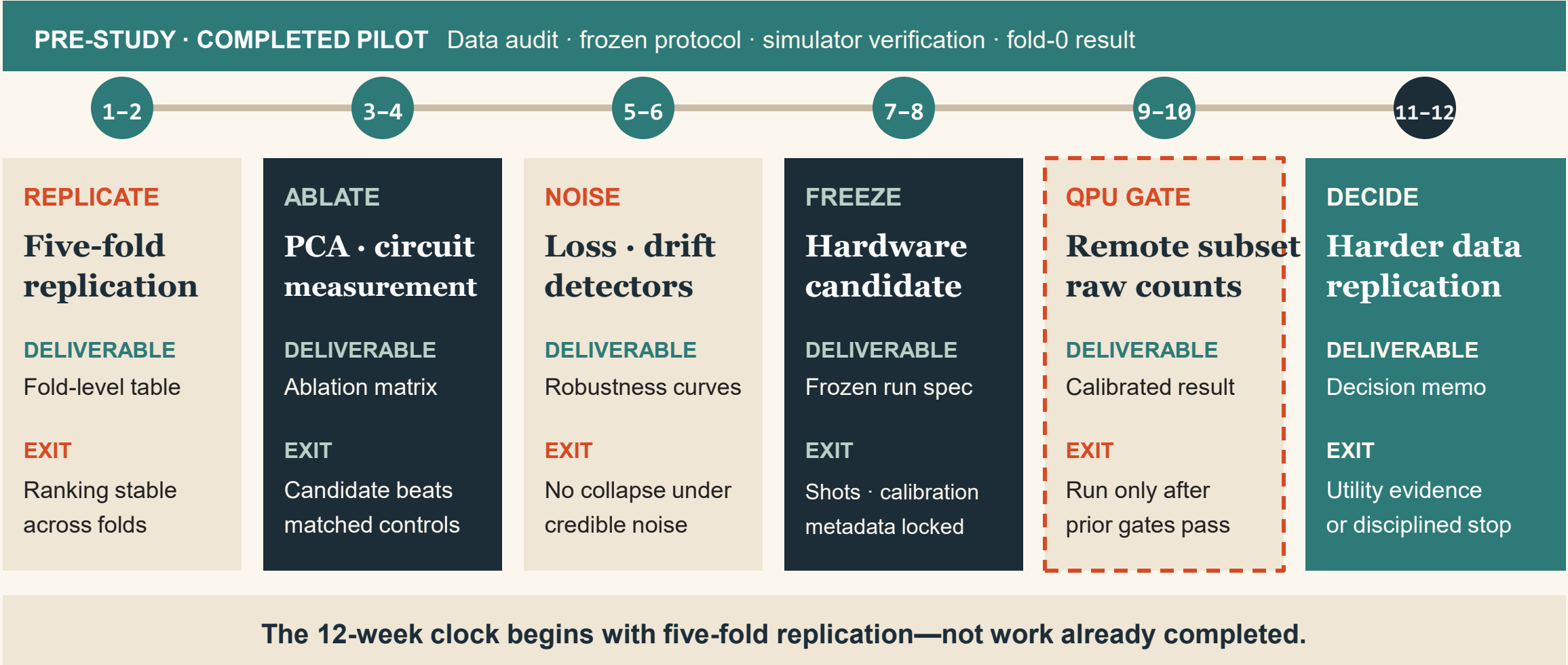
**Close ELM / RBF gap
across outer folds**
or show practical resource value

STOP GATE

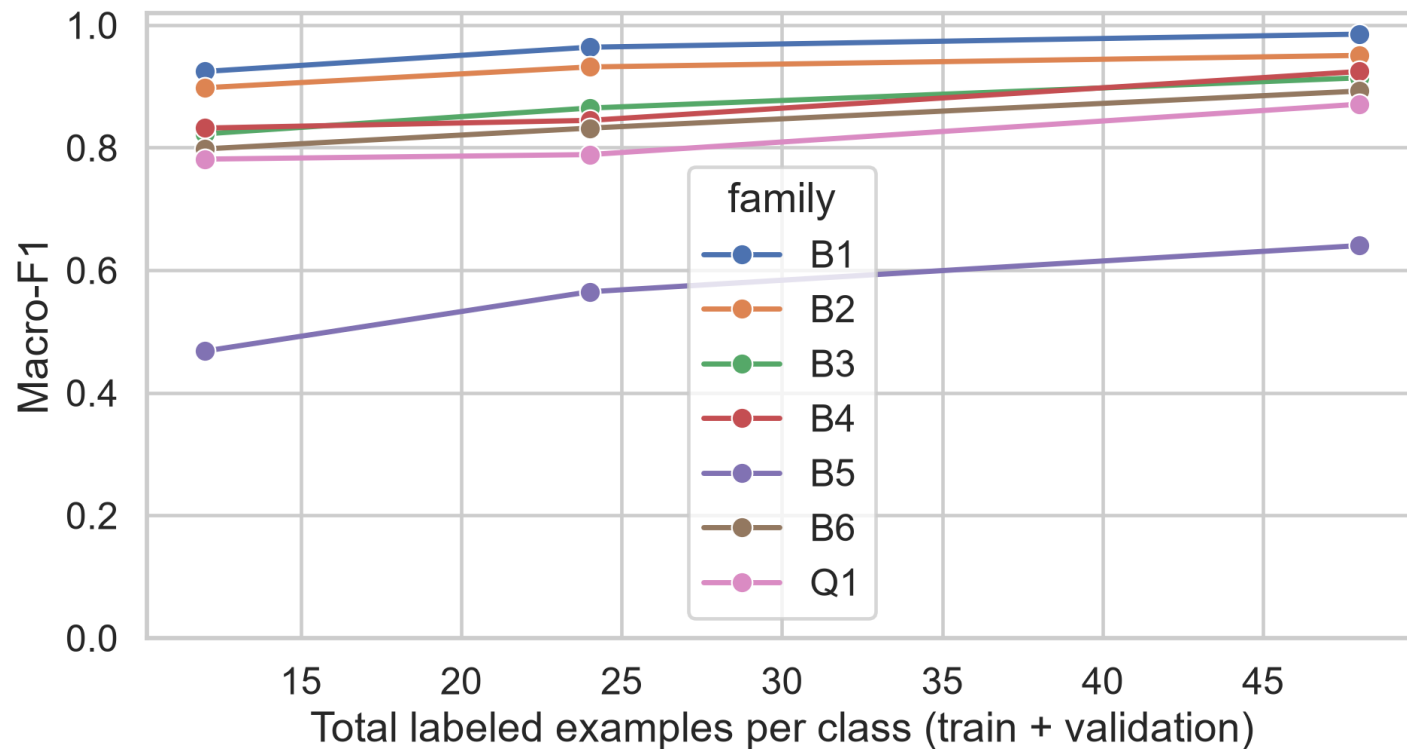
**If Q1 only beats RFF
while strong controls win
FEASIBILITY WITHOUT UTILITY**

Source: frozen experiment protocol · repeated evaluations share one outer test fold

The 12-week project starts after the completed pilot



The pilot shows useful signal—not competitive advantage



Q1 · PHOTONIC-15

0.7812 → 0.7886

→ 0.8708

12 · 24 · 48 labels per class

B1 · FULL CLIP + LINEAR

0.9242 → 0.9638

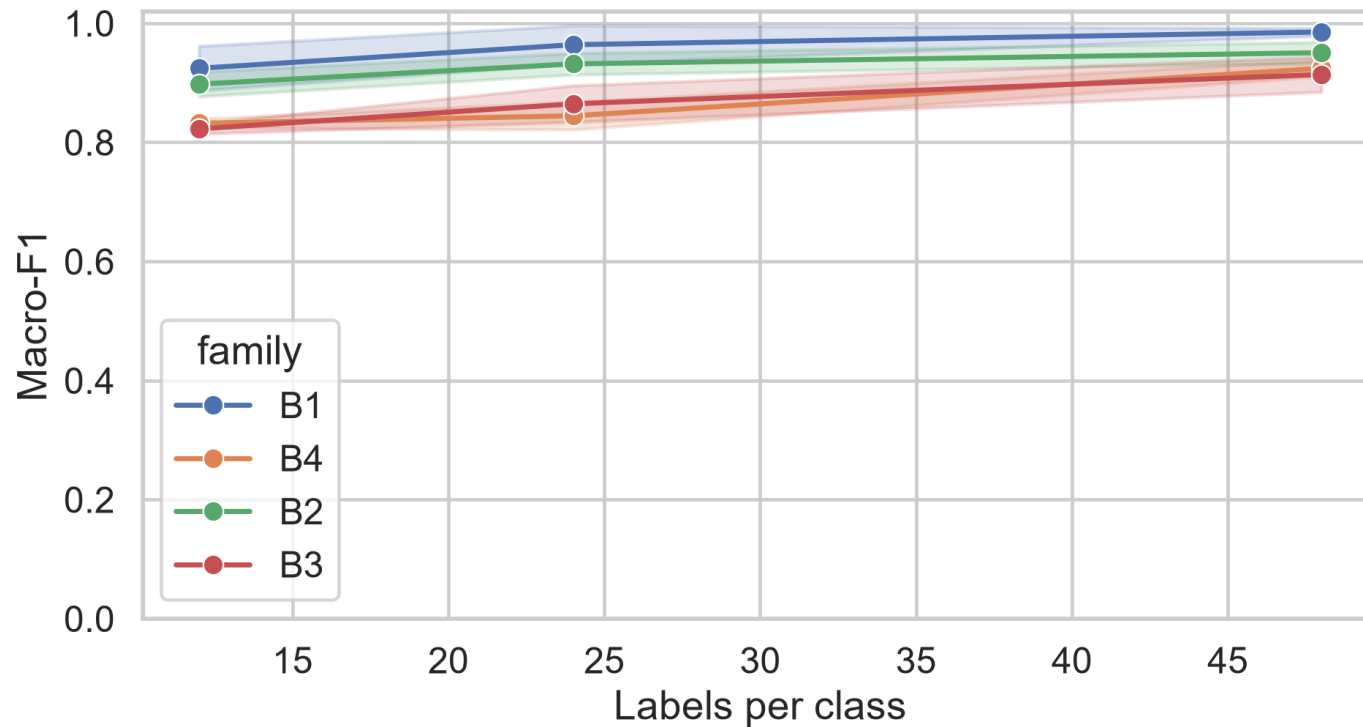
→ 0.9852

PILOT VERDICT

Learns signal;
not competitive yet.

Mean ± sample SD across three subset seeds · one outer fold · source: main_results.csv

Compression—not photonics alone—explains the shortfall



FULL → PCA-5 PENALTY · B1 – B3

+.1018 · +.0993 · +.0713

budgets 12 · 24 · 48

Q1 – RFF

+.3133 · +.2238 · +.2307

Q1 – ELM

–.0166 · –.0432 · –.0215

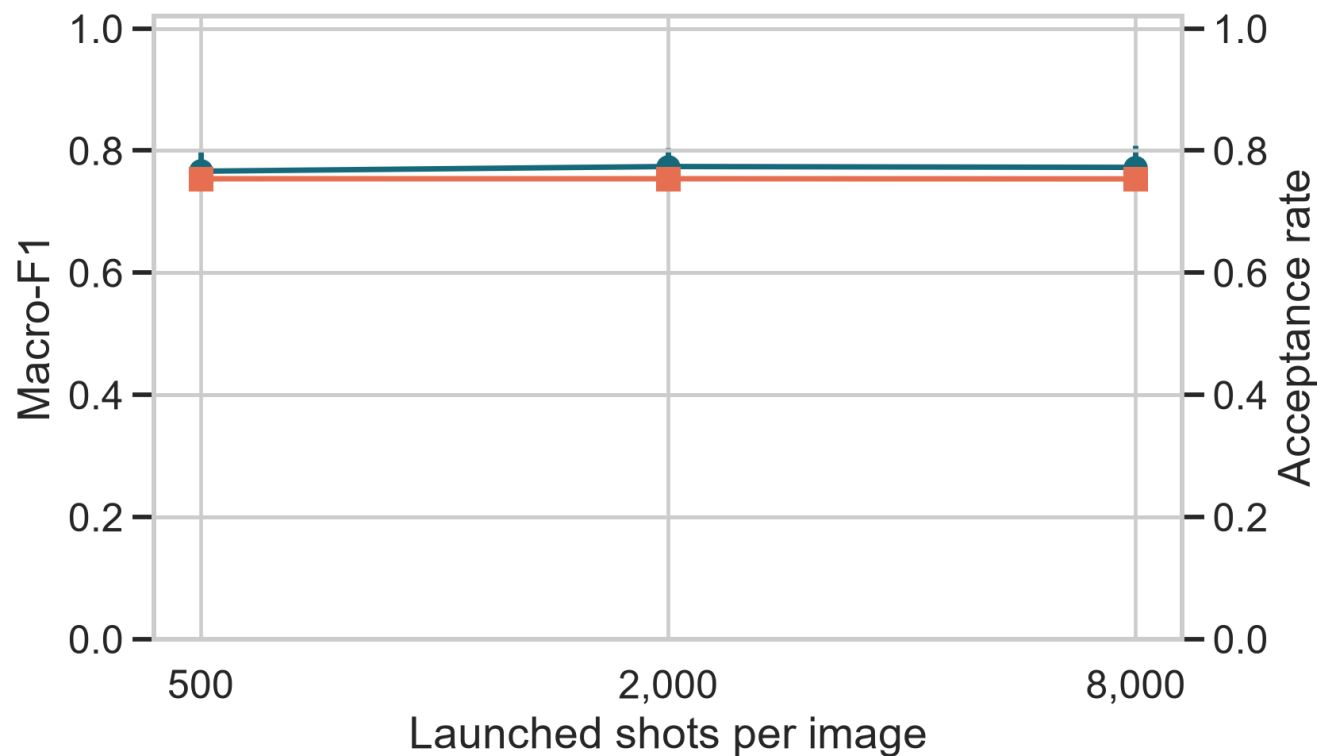
Q1 – PCA-5 RBF

–.0507 · –.0558 · –.0534

Q1 only wins against the weak matched map.

Separate information lost in PCA-5 from the value added by the photonic circuit.

Finite-shot sampling is stable; accuracy does not recover



MACRO-F1 · 15 EVALUATIONS EACH

500 $0.7662 \pm .0322$

2,000 $0.7739 \pm .0308$

8,000 $0.7725 \pm .0363$

acceptance ≈ 0.754

0 zero-acceptance failures

TESTED

Shot sampling
Acceptance
Ideal-trained readout

NOT YET TESTED

Loss · phase drift
Distinguishability
Detectors · noise training

Technically feasible; practical utility remains open

PROVEN

Technical feasibility

- Executable pipeline
- Simulator semantics
- Reproducible artifacts

PARTIALLY ANSWERED

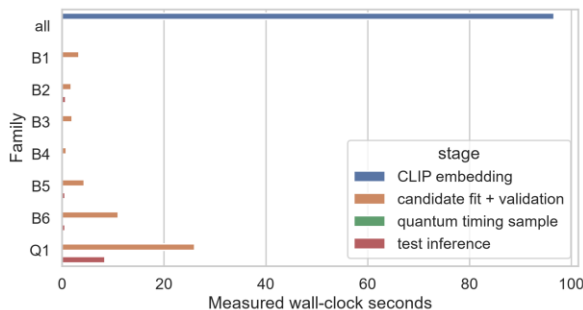
Application signal

- Learning with labels
- Paired one-fold ranking
- Finite-shot sampling

UNANSWERED

Practical utility

- Five-fold generalization
- Realistic photonic noise
- Remote hardware execution



MEASURED ON CPU

OpenCLIP embed. **96.57 s**

R1 models **≈59.9 s**

Runtime permits broader validation.

DECISION CONSTRAINT

Scientific validity

- One outer fold · small validation cells
- ideal optics · B1 near ceiling
- RUNTIME IS NOT THE BOTTLENECK**

Local CPU profile · no loss, phase drift, detector noise, or QPU queue

Continue only through a decision-changing test

WHICH APPLICATION?

Few-label industrial defect inspection

with a fixed photonic reservoir

WHY THIS IDEA?

Real label scarcity · compact phase encoding

measurable evidence within 12 weeks

CURRENT VERDICT

**Technically feasible and
scientifically interesting**

NO DEMONSTRATED PRACTICAL ADVANTAGE YET

CONTINUE ONLY IF

**Q1 closes the ELM / RBF gap
across outer folds**

or shows a practical resource trade-off.

STOP CONDITION

**If Q1 only beats RFF:
report feasibility without utility.**

Evidence first · hardware only after gates pass

Repository · github.com/MonsieurNam/Quanova_Photonic_Defect_QML